

1 Solution — GIAM 4.3

Problem 1

1.1 Problem

Let $A = \{1, 2, \{1, 2\}, b\}$ and let $B = \{a, b, \{1, 2\}\}$. Find the following:

- a. $A \cap B$
- b. $A \cup B$
- c. $A \setminus B$
- d. $B \setminus A$
- e. $A \triangle B$

1.2 Answer

- a. $A \cap B = \{b, \{1, 2\}\}$
- b. $A \cup B = \{1, 2, a, b, \{1, 2\}\}$
- c. $A \setminus B = \{1, 2\}$
- d. $B \setminus A = \{a\}$
- e. $A \triangle B = \{1, 2, a\}$

All five were computed independently and agree.

1.3 Working

There are five distinct objects in play altogether: the numbers **1** and **2**, the letters **a** and **b**, and the **set** $\{1, 2\}$. Every question below reduces to asking, for each of these five, which of A and B it belongs to.

object	$\in A?$	$\in B?$
1	yes	no
2	yes	no
a	no	yes
b	yes	yes
$\{1, 2\}$	yes	yes

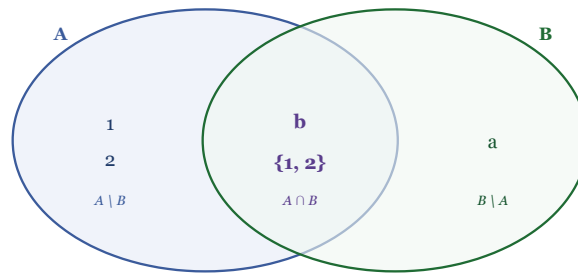
Reading the table off gives everything at once. The objects with **yes, yes** form $A \cap B$; those with at least one yes form $A \cup B$; **yes, no** gives $A \setminus B$; **no, yes** gives $B \setminus A$; and the two mixed rows together give the symmetric difference.

1.4 The Picture

THE TWO SETS

$$A = \{1, 2, \{1, 2\}, b\} \quad B = \{a, b, \{1, 2\}\}$$

A has four elements, B has three. Their union has five — every membership test below asks only whether one whole object appears in the other list.



Five distinct objects in all. The object $\{1, 2\}$ sits in the overlap; the numbers 1 and 2 do not.

PART	EXPRESSION	VALUE	SIZE
a	$A \cap B$	$\{b, \{1, 2\}\}$	2
b	$A \cup B$	$\{1, 2, a, b, \{1, 2\}\}$	5
c	$A \setminus B$	$\{1, 2\}$	2
d	$B \setminus A$	$\{a\}$	1
e	$A \triangle B$	$\{1, 2, a\}$	3

THE TRAP: $\{1, 2\}$ IS ONE OBJECT, NOT TWO

$\{1, 2\} \in A$ and $\{1, 2\} \in B$, so it survives the intersection. But $1 \notin B$ and $2 \notin B$, so the numbers 1 and 2 fall out of $A \cap B$ and into $A \setminus B$. All four memberships checked directly.

Curiosity: $\{1, 2\}$ is simultaneously an *element* of A and a *subset* of A — but only an element of B, since 1 and 2 are missing there. Exactly the $\in$ versus $\subset$ distinction of Section 2.

THE THREE-PIECE PARTITION

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A)$$

$$5 = 2 + 2 + 1$$

The three pieces are pairwise disjoint, so parts a, c and d already determine b and e. Verified.

Figure 1.1: A diagram in four parts. The first states the two sets: A equals the set containing 1, 2, the set containing 1 and 2, and b; B equals the set

containing a , b , and the set containing 1 and 2. A has four elements, B has three, and their union has five, since every membership test asks only whether one whole object appears in the other list. The second part is a Venn diagram of two overlapping ovals, A in blue on the left and B in green on the right. In the left crescent, labelled A minus B , sit the numbers 1 and 2. In the overlap, labelled A intersect B , sit b and the set containing 1 and 2. In the right crescent, labelled B minus A , sits a . A caption notes there are five distinct objects in all, and that the object consisting of the set containing 1 and 2 sits in the overlap while the numbers 1 and 2 do not. The third part is a table of the five answers with their sizes: A intersect B is b and the set containing 1 and 2, of size 2; A union B is 1, 2, a , b and the set containing 1 and 2, of size 5; A minus B is 1 and 2, of size 2; B minus A is a , of size 1; and the symmetric difference is 1, 2 and a , of size 3. A red band below explains the trap, that the set containing 1 and 2 is one object and not two: it belongs to both A and B and so survives the intersection, but 1 and 2 themselves do not belong to B , so they fall out of the intersection and into A minus B , with all four memberships checked directly. It adds that the set containing 1 and 2 is simultaneously an element of A and a subset of A , but only an element of B , since 1 and 2 are missing there, which is exactly the element versus subset distinction of Section 2. A green band at the bottom gives the three-piece partition: A union B equals A minus B , together with A intersect B , together with B minus A , with sizes 5 equals 2 plus 2 plus 1. The three pieces are pairwise disjoint, so parts a , c and d already determine parts b and e , as verified.

1.5 Remark — The Trap: $\{1, 2\}$ Is One Object, Not Two

The whole point of the exercise is the row $\{1, 2\}$ in the table above. Braces inside a set are not decoration: $\{1, 2\}$ is a single element of A , sitting alongside 1, 2 and b , which is why

$$|A| = 4 \quad \text{and not} \quad |A| = 5.$$

Since that one object is listed in both A and B , it survives into the intersection. Meanwhile the *numbers* 1 and 2 appear nowhere in B — checked directly: $1 \notin B$ and $2 \notin B$ — so they fall out of $A \cap B$ and into $A \setminus B$. The set $\{1, 2\}$ and the numbers it contains travel in opposite directions.

A pleasant curiosity sits in the same data. Here

$$\{1, 2\} \in A \quad \text{and} \quad \{1, 2\} \subseteq A,$$

both true at once — the first because $\{1, 2\}$ is listed, the second because 1 and 2 are also listed. For B only the first holds, since $1 \notin B$. So the same pair of symbols behaves differently against the two sets, which is precisely the $\in$ against $\subseteq$ distinction drilled in Section 4.2, and A is an instance of the phenomenon studied in 4.2 Problem 5.

1.6 Remark — The Three-Piece Partition

The five answers are not independent. Any two sets cut their union into three pairwise disjoint pieces:

$$A \cup B = (A \setminus B) \cup (A \cap B) \cup (B \setminus A),$$

verified here, with $(A \setminus B) \cap (B \setminus A) = \emptyset$. Counting the pieces gives

$$5 = 2 + 2 + 1.$$

So parts **a**, **c** and **d** already determine the rest:

$$\begin{aligned} A \cup B &= (A \setminus B) \cup (A \cap B) \cup (B \setminus A), \\ A \triangle B &= (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B), \end{aligned}$$

both forms of the symmetric difference confirmed to give $\{1, 2, a\}$. Collapsing the partition count yields inclusion–exclusion,

$$|A \cup B| = |A| + |B| - |A \cap B|, \quad 5 = 4 + 3 - 2,$$

the subtraction correcting for the overlap having been counted twice — a quick arithmetic check on any answer of this shape.

1.7 Remark — Everything Above Is Bitwise Arithmetic

List the five objects of $A \cup B$ in a fixed order and record each set by which slots it occupies:

order	1, 2, a , b , $\{1, 2\}$
A	11011
B	00111

The five operations then become the five familiar bitwise ones, all verified to reproduce the answers above:

set operation	bitwise	result
$A \cap B$	AND	00011
$A \cup B$	OR	11111
$A \setminus B$	A AND NOT B	11000
$B \setminus A$	B AND NOT A	00100
$A \triangle B$	XOR	11100

This is how sets are represented inside a computer, and it is why the symmetric difference — the odd one out in most textbooks — is really the most natural of the five: it is plain XOR.

1.8 Remark — Symmetric Difference Makes the Power Set a Group

Fix the universe $U = A \cup B$, so $\mathcal{P}(U)$ has $2^5 = 32$ members. Exhaustive check over all of them — and all 32^3 triples for associativity — confirms:

$$\begin{array}{ll}
 X \triangle \emptyset = X & \text{(identity)} \\
 X \triangle X = \emptyset & \text{(every element is its own inverse)} \\
 (X \triangle Y) \triangle Z = X \triangle (Y \triangle Z) & \text{(associativity)} \\
 X \triangle Y = Y \triangle X & \text{(commutativity)}
 \end{array}$$

So $(\mathcal{P}(U), \triangle)$ is an abelian group of order 32 in which every non-identity element has order 2 — the elementary abelian group $(\mathbb{Z}/2\mathbb{Z})^5$. Neither $\cup$ nor $\cap$ manages this, since neither has inverses: there is no X with $A \cup X = \emptyset$.

Adding intersection gives more. Distributivity

$$X \cap (Y \triangle Z) = (X \cap Y) \triangle (X \cap Z)$$

was verified over all 32^3 triples, so $(\mathcal{P}(U), \triangle, \cap)$ is a commutative ring with unit U in which every element is idempotent — a **Boolean ring**, and by Stone's representation theorem every such ring arises from a field of sets in essentially this way.